



APUNTES DE CLASE

IF-6008 TÓPICOS DE ASTRONOMÍA Y ASTROFÍSICA

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Licenciatura en Ingeniería Física

Escuela de Física

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1 Cosmología y Galaxias

¿De qué está hecho el universo? ¿Cómo comenzo? ¿Cómo ha evolucionado durante los 13.800 millones de años desde su origen? ¿Y cómo terminará? Estas son las preguntas que aborda la cosmología, el estudio del universo en su conjunto. La investigación en cosmología involucra astronomía, pero también física gravitacional, física de partículas y preguntas desafiantes sobre la interpretación de fenómenos que no podemos ver directamente, como la posible existencia de algo antes del Big Bang.

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2 Teoría del Big Bang

3 Ley de Hubble

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu} - g_{\mu\nu}\Lambda \quad (1)$$

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + pg_{\mu\nu} \quad (2)$$

$$\rho = \sum \rho_i = \rho_m + \rho_r + \rho_\Lambda + \rho_\kappa \quad (3)$$

$$ds^2 = -dt^2 + a^2(t) \left[\frac{dr^2}{1 - \kappa r^2} + r^2 d\vartheta^2 + r^2 \sin^2 \vartheta d\varphi^2 \right] \quad (4)$$

$$\frac{\dot{a}^2}{a^2} + \frac{\kappa c^2}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda c^2}{3} \quad (5)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) + \frac{\Lambda c^2}{3} \quad (6)$$

$$p = p(\rho)$$

$$p_i = \omega_i \rho_i c^2 \quad (7)$$

$$\dot{\rho}_i + 3H(1 + \omega_i)\rho_i \quad (8)$$

$$H \equiv H(t) \equiv \frac{da}{dt} \frac{1}{a} \quad (9)$$

$$H_0 = H(t_0) = 1$$

$$H_0 = 100 h \text{ km s}^{-1} \text{ Mpc}^{-1}$$

?

$$\rho_c \equiv \frac{3H^2}{8\pi G} \quad (10)$$

$$\Omega_i(t) \equiv \frac{\rho_i(t)}{\rho_c(t)} \quad (11)$$

$$\Omega = \Omega_m + \Omega_r + \Omega_\Lambda + \Omega_\kappa \quad (12)$$

4 Formación galáctica

$$\rho(\mathbf{x}, t) = \bar{\rho}(t)[1 + \delta(\mathbf{x}, t)] \quad (13)$$

$$p(\mathbf{x}, t) = \bar{p}(t)[1 + \delta(\mathbf{x}, t)] \quad (14)$$

$$\frac{\partial^2 \delta}{\partial t^2} + 2H \frac{\partial \delta}{\partial t} = 4\pi G \bar{\rho} \delta - \frac{c_s^2}{a^2} \nabla^2 \delta \quad (15)$$

$$c_s^2 \equiv \frac{\partial p}{\partial \rho} \quad (16)$$

$$\delta(\mathbf{x}, t) = \sum_{\mathbf{k}} \delta_{\mathbf{k}}(t) e^{i\mathbf{k} \cdot \mathbf{x}} \quad (17)$$

$$\ddot{\delta}_{\mathbf{k}} + 2H \dot{\delta}_{\mathbf{k}} = -\omega_{\mathbf{k}}^2 \delta_{\mathbf{k}} \quad (18)$$

$$\omega_{\mathbf{k}}^2 = \left(\frac{kc_s}{H} \right)^2 - 4\pi G \bar{\rho} \quad (19)$$

$$k \equiv |\mathbf{k}| = 2\pi a / \lambda$$

$$\lambda_J \equiv \frac{2\pi a}{k_J} = c_s \sqrt{\frac{\pi}{G \bar{\rho}}} \quad (20)$$

5 Tipos de galaxias

Peebles [2021]

Bag and Liivamägi [2021]

Khetan, Nandita et al. [2021]

6 Enlaces de interés



European Southern Observatory

<https://www.eso.org/>

The Sloan Digital Sky Survey

<https://www.sdss.org/>

Center for Astrophysics - Harvard & Smithsonian

<https://www.cfa.harvard.edu/research/science-field/cosmology>

Webb Space Telescope

<https://www.jwst.nasa.gov/>

Astropy

<https://www.astropy.org/>

SciServer

<https://www.sciserver.org/>
[Taghizadeh-Popp et al., 2020]

The Dark Energy Survey

<https://www.darkenergysurvey.org/>

Marvin - the ultimate tool to visualise and analyse MaNGA data

<https://dr17.sdss.org/marvin/>

Hubble Space Telescope

<https://hubblesite.org/>

The SAO/NASA Astrophysics Data System

<https://ui.adsabs.harvard.edu/>

Centre de Données astronomiques de Strasbourg - Strasbourg astronomical Data Center

<https://cds.u-strasbg.fr/>
<http://simbad.u-strasbg.fr/simbad/> Wenger, M. et al. [2000]

SAOImageDS9 - An image display and visualization tool for astronomical data

<https://sites.google.com/cfa.harvard.edu/saoimageds9/>

The Millennium simulation

<https://wwwmpa.mpa-garching.mpg.de/millennium/>

The Stephen Hawking Centre for Theoretical Cosmology

<https://www.etc.cam.ac.uk/>

Astronomy and Computing

<https://www.sciencedirect.com/journal/astronomy-and-computing>

Astronomy and Astrophysics

<https://www.aanda.org/>

Annual Review of Astronomy and Astrophysics

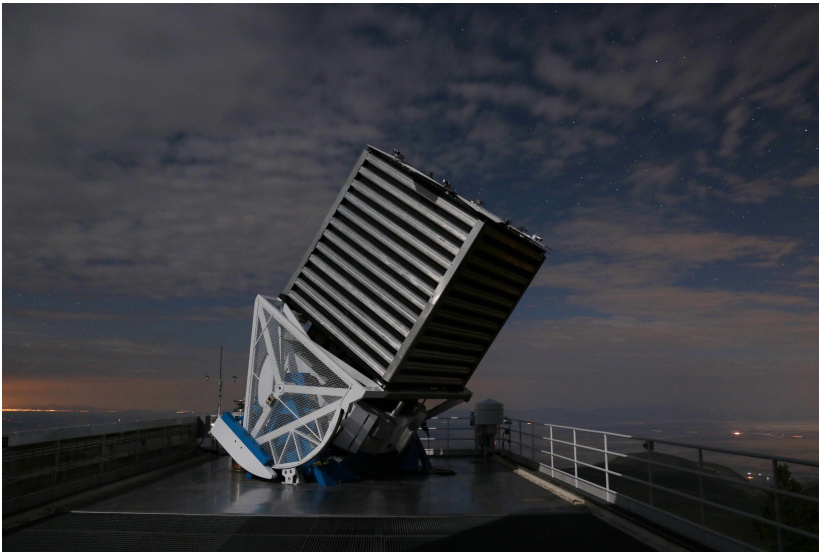
<https://www.annualreviews.org/journal/astro>

Monthly Notices of the Royal Astronomy Society

<https://academic.oup.com/mnras>



Figure 1: Vía Láctea. Créditos G. Hüdepohl (atacamaphoto.com)/ESO



$1 \text{ Mpc} = 10^6 \text{ pc} = 3.086 \times 10^{22} \text{ m}$.

References

- S. Bag and L. J. Liivamägi. The shape distribution of superclusters in sdss dr 12, 2021.
- Khetan, Nandita, Izzo, Luca, Branchesi, Marica, Wojtak, Radoslaw, Cantiello, Michele, Murugesan, Chandrashekar, Agnello, Adriano, Cappellaro, Enrico, Della Valle, Mas-

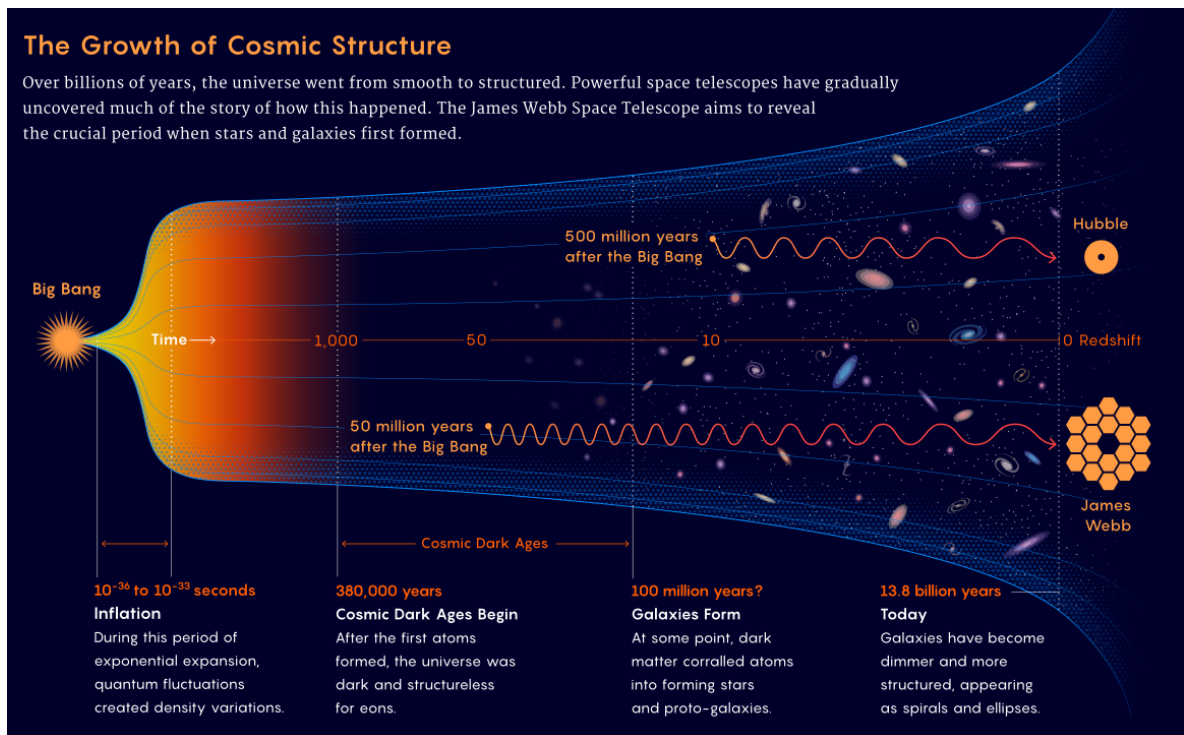


Figure 2: Vía Láctea. Créditos: Samuel Velasco ([Quanta Magazine](#))

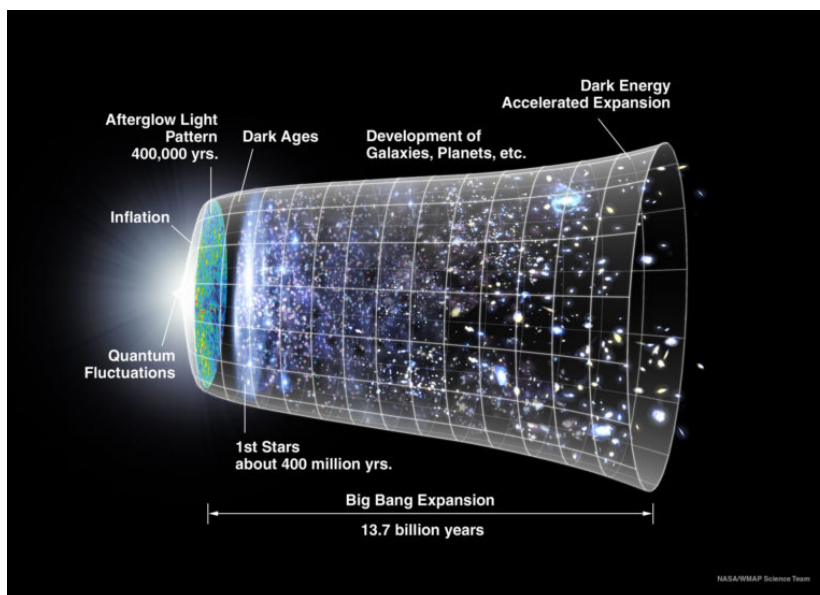


Figure 3: Timeline. Créditos G. Hüdepohl ([atacamaphoto.com](#))/ESO

simo, Gall, Christa, Hjorth, Jens, Benetti, Stefano, Brocato, Enzo, Burke, Jamison, Hiramatsu, Daichi, Howell, D. Andrew, Tomasella, Lina, and Valenti, Stefano. A new measurement of the hubble constant using type ia supernovae calibrated with surface brightness fluctuations. *A&A*, 647:A72, 2021. doi: 10.1051/0004-6361/202039196. URL <https://doi.org/10.1051/0004-6361/202039196>.

P. J. E. Peebles. The extended local supercluster, 2021.

M. Taghizadeh-Popp, J. Kim, G. Lemson, D. Medvedev, M. Raddick, A. Szalay, A. Thakar, J. Booker, C. Chhetri, L. Dobos, and et al. Sciserver: A science platform for astronomy and beyond. *Astronomy and Computing*, 33:100412, Oct 2020. ISSN 2213-1337. doi: 10.1016/j.ascom.2020.100412. URL <http://dx.doi.org/10.1016/j.ascom.2020.100412>.

Wenger, M., Ochsenbein, F., Egret, D., Dubois, P., Bonnarel, F., Borde, S., Genova, F., Jasniewicz, G., Laloë, S., Lesteven, S., and Monier, R. The simbad astronomical database - the cds reference database for astronomical objects. *Astron. Astrophys. Suppl. Ser.*, 143 (1):9–22, 2000. doi: 10.1051/aas:2000332. URL <https://doi.org/10.1051/aas:2000332>.